

# A SHARP DICHOTOMY FOR BOUNDED-GAP SUMSETS

JOHAN LAND

ABSTRACT. Let  $k \geq 3$  and  $1 \leq d_1 < d_2$ . We prove that Erdős Problem 1112 has an affirmative answer exactly when  $d_2 \geq k + 1$ ; in that range the explicit lacunarity ratio  $d_2 + 2$  works. Below the threshold we prove more: however rapidly prescribed successive ratios grow, one sequence meets the  $k$ -fold sumset of every admissible bounded-gap sequence. The proof combines a reciprocal Beatty construction, Kneser's density theorem, and an elementary proof of the sharp finite subset-sum lemma required at the boundary.

For a set  $A \subset \mathbb{N}$ , write

$$kA := \{a_1 + \cdots + a_k : a_1, \dots, a_k \in A\},$$

with repetitions allowed. A sequence  $A = \{a_1 < a_2 < \cdots\}$  is  $(d_1, d_2)$ -admissible if  $d_1 \leq a_{n+1} - a_n \leq d_2$  for every  $n$ . We address the formulation recorded as Erdős Problem 1112 [1], inspired by a remark of Erdős and Graham [2]. The earlier proof [5] established the dichotomy with the explicit bound  $192d_2$  and was formally verified in Lean. The present reconstruction improves the positive bound and replaces the finite certificate layer with an elementary proof. The verification scope is recorded in Appendix B.

**Theorem 1** (Sharp dichotomy). *Let  $k \geq 3$  and  $1 \leq d_1 < d_2$  be integers.*

- (1) *If  $d_2 \geq k + 1$ , every sequence of positive integers  $B = \{b_1 < b_2 < \cdots\}$  with  $b_{n+1} \geq (d_2 + 2)b_n$  admits a  $(d_1, d_2)$ -admissible  $A$  such that  $kA \cap B = \emptyset$ .*
- (2) *If  $d_2 \leq k$ , no fixed ratio works. More strongly, for every sequence  $R_1, R_2, \dots$  of positive integer lower bounds there is one increasing sequence  $B$  of positive integers with  $b_{n+1} \geq R_n b_n$  which meets  $kA$  for every  $(d_1, d_2)$ -admissible  $A$ .*

Thus the answer is yes exactly when  $d_2 \geq k + 1$ .

Above the threshold, a reciprocal Beatty sequence avoids a prescribed lacunary sequence. Below it, every admissible walk has a congruence-class tail in its  $k$ -fold sumset; a diagonal construction then meets every such sumset. The sharp finite input is Theorem 2.2, proved in the appendix.

## 1. THE EXISTENCE CONSTRUCTION

Put  $D = d_2$  and assume  $D \geq k + 1$ .

**Lemma 1.1** (Lacunary interpolation). *Let  $I = [u, v] \subset (0, 1)$  have length  $\ell > 0$ , let  $J$  be a nondegenerate closed bounded real interval, and let positive integers  $q_1 < q_2 < \cdots$  satisfy  $q_{n+1} \geq \rho q_n$ . If*

$$\ell(\rho - 1) \geq 1, \tag{1}$$

*then, after deleting finitely many  $q_n$ , some  $\theta \in J$  satisfies  $\{q_n \theta\} \in I$  for every remaining  $n$ .*

*Proof.* For a positive integer  $q$ , the set

$$E_q = \{\theta : \{q\theta\} \in I\} = \bigcup_{m \in \mathbb{Z}} [(m + u)/q, (m + v)/q]$$

has components of length  $\ell/q$ , with left endpoints spaced by  $1/q$ . For some  $N$ , the interval  $[q_N \inf J - u, q_N \sup J - v]$  has length  $q_N |J| - \ell \geq 1$ , so it contains an integer. Thus a component  $K_N$  of  $E_{q_N}$  lies in  $J$ . If  $K_n$  is a component of  $E_{q_n}$ , the allowable indices of components of  $E_{q_{n+1}}$  contained in  $K_n$  form a real interval of length

$$q_{n+1}|K_n| - \ell = \ell(q_{n+1}/q_n - 1) \geq 1.$$

It contains an integer, giving  $K_{n+1} \subseteq K_n$ . The nested nonempty compact intervals have a common point  $\theta$ , as required.  $\square$

**Proposition 1.2** (Existence). *If  $D \geq k + 1$ , the ratio  $D + 2$  works.*

*Proof.* Given  $B$  with  $b_{n+1} \geq (D + 2)b_n$ , put

$$\varepsilon = \frac{1}{2D(D+1)}, \quad I = [\varepsilon, \varepsilon + 1/(D+1)], \quad J = [1/D, 1/D + \varepsilon/(D-1)].$$

Lemma 1.1 applies with equality in (1). It yields  $\theta \in J$  and  $N_0$  such that

$$\{b_n \theta\} \in I \quad (n \geq N_0). \quad (2)$$

Put  $\alpha = 1/\theta$ . Then

$$D - 1 < \alpha \leq D, \quad (3)$$

and, since  $k \leq D - 1$ ,

$$1 - k\theta \geq 1 - (D - 1) \max J = 1/D - \varepsilon = \varepsilon + 1/(D + 1) = \max I. \quad (4)$$

Choose an integer  $m \geq 1$  with  $k \lfloor \alpha m \rfloor > \max_{n < N_0} b_n$ , reading the empty maximum as zero, and set

$$A = \{\lfloor \alpha(m + i) \rfloor : i = 0, 1, 2, \dots\}.$$

Its gaps are  $D - 1$  or  $D$ , hence admissible, and its least  $k$ -fold sum exceeds every discarded term. If a remaining  $b$  were a sum of  $k$  such floors, then for an integer  $S$  (the sum of their indices)  $b \leq \alpha S < b + k$ , whence

$$b\theta \leq S < (b + k)\theta.$$

But  $0 < \{b\theta\} \leq 1 - k\theta$  by (2)–(4), so this half-open interval contains no integer. This is a contradiction.  $\square$

*Remark 1.3.* The argument works for every real ratio  $\rho > 1 + D/(D - k)$ : take  $J$  sufficiently close to  $1/D$  and a closed interval  $I \subset (0, 1 - k \sup J)$  of length  $1/(\rho - 1)$ .

## 2. TOOLS FOR TAIL COVERING

Call  $A$  *k-tail-covering* if, for integers  $q \geq 1$ ,  $0 \leq r < q$ , and  $X_0$ ,

$$\{x \geq X_0 : x \equiv r \pmod{q}\} \subseteq kA. \quad (5)$$

**2.1. The density shortcut and the finite input.** For  $X \subseteq \mathbb{N}_0$  write  $\underline{d}(X) = \liminf_{N \rightarrow \infty} |X \cap [0, N]|/(N + 1)$ .

**Lemma 2.1** (Kneser shortcut). *Let  $P = \{p_0 < p_1 < \dots\}$  with  $p_0 = 0$ . If  $p_n \leq cn + C$  eventually, for constants  $0 < c < k$  and  $C$ , then  $kP$  contains a congruence-class tail.*

*Proof.* For large  $N$ , take  $n = \lfloor (N - C)/c \rfloor$ . Then  $p_n \leq N$ , so  $\underline{d}(P) \geq 1/c > 1/k$ . The lower-density consequence of Kneser's theorem [4, 3] needed here is that either  $\underline{d}(kP) \geq k\underline{d}(P)$  or  $kP$  contains a congruence-class tail. The first alternative is impossible because density is at most one, proving the claim.  $\square$

For a finite multiset  $S$ , write  $\text{SS}(S) = \{\sum_{x \in T} x : T \text{ is a submultiset of } S\}$ .

**Theorem 2.2** (Finite interval lemma (SHARP)). *If  $G$  is a finite set of positive integers with  $|G| \geq 3$ ,  $\gcd(G) = 1$ , and  $M = \max G$ , some multiset of at most  $M - 1$  elements of  $G$  has  $M$  consecutive subset sums.*

The elementary proof in Appendix A uses no search or certificate list.

**2.2. A slot lemma.** For a gap word  $g_1, g_2, \dots$ , put  $p_0 = 0$  and  $p_n = g_1 + \dots + g_n$ . A tail of the walk is a translate of  $P = \{p_n : n \geq 0\}$ .

**Lemma 2.3** (Slots). *Suppose all gaps belong to a finite  $G$ , every member recurs infinitely often,  $|G| \geq 3$ ,  $\gcd(G) = 1$ , and  $M = \max G \leq k$ . Then  $kP$  contains every sufficiently large integer.*

*Proof.* By SHARP choose  $S = \{\delta_1, \dots, \delta_m\}$ ,  $m \leq M - 1 \leq k - 1$ , with  $[c, c + M - 1] \subseteq \text{SS}(S)$ . Choose distinct occurrences  $g_{n_j} = \delta_j$ . In summand slot  $j$  choose  $p_{n_j-1}$  or  $p_{n_j}$ . These slots realize a fixed base plus every offset in  $[c, c + M - 1]$ . Park  $k - 1 - m$  summands at  $p_0 = 0$ . The last summand is a dial  $p_n$ . For a large target  $x$ , the residual  $x - \text{base} - p_n$  decreases to  $-\infty$  in jumps at most  $M$ . Its first value at most  $c + M - 1$  is at least  $c$ , so the slots supply the residual. This represents  $x$  in  $kP$ .  $\square$

## 3. EVERY SUBCRITICAL WALK IS TAIL-COVERING

**Proposition 3.1.** *For  $k \geq 3$ , every increasing sequence of positive integers with gaps at most  $k$  is  $k$ -tail-covering.*

*Proof.* Discard a finite prefix so that the gaps lie in the nonempty tail alphabet  $G$  of values recurring infinitely often. Put  $p_0 = 0$  and  $p_n = g_1 + \cdots + g_n$ . Divide all gaps and prefix sums by  $g = \gcd(G)$ . A class tail modulo  $g$  rescales to one modulo  $gg$ ; translating back only changes the residue. Relabel so that  $\gcd(G) = 1$ . If  $M = \max G < k$ , then  $p_n \leq Mn$ , and Lemma 2.1 applies. Thus assume  $M = k$ . If  $|G| \geq 3$ , use Lemma 2.3. The remaining alphabet is  $G = \{\delta, k\}$  with  $1 \leq \delta < k$  and  $\gcd(\delta, k) = 1$ . Set  $e = k - \delta$ , encode gap  $k$  by  $h_n = 1$  and gap  $\delta$  by  $h_n = 0$ , and write  $q_n = h_1 + \cdots + h_n$ . Then

$$p_n = \delta n + e q_n. \quad (6)$$

For  $s \geq 0$  set

$$W_s = \left\{ \sum_{j=1}^k q_{i_j} : i_j \geq 0, \sum_{j=1}^k i_j = s \right\}.$$

This is an integer interval  $[w^-(s), w^+(s)] \cap \mathbb{Z}$ : compositions are connected by unit transfers, each changing the value by  $-1, 0$ , or  $1$ . A path between extremizers misses no intermediate value. Adding or deleting one unit shows that each endpoint rises by  $0$  or  $1$  as  $s$  increases by one. Write  $\text{width}(s) = w^+(s) - w^-(s)$ .

*Sweep.* If  $\text{width}(s) \geq k - 1$  eventually, then  $kP$  contains every large integer. Indeed, a target  $x$  is represented at budget  $s$  exactly when  $w(s) = (x - \delta s)/e \in W_s$ . The integral values of  $w(s)$  occur in one class of  $s$  modulo  $e$ . Along this class  $w$  drops by  $\delta$  per step, whereas either endpoint rises by at most  $e$ . Since  $w^+(s) \leq s$ , for large  $x$  the target starts above the band at the first admissible budget beyond the width threshold; eventually it becomes negative. If it skips the band, consecutive admissible budgets satisfy

$$w^+(s) < w(s) = w(s + e) + \delta \leq w^-(s + e) - 1 + \delta \leq w^-(s) + e - 1 + \delta.$$

Thus  $\text{width}(s) \leq k - 2$ , a contradiction.

A binary word is *balanced* if equal-length factors differ in their numbers of ones by at most one. Define

$$V(\sigma) = \max_{i+j=\sigma} (q_i + q_j) - \min_{i+j=\sigma} (q_i + q_j).$$

If  $V(\sigma) \leq 1$  for every  $\sigma$ , the word is balanced: comparing factors with endpoint indices  $i, i'$  and  $j', j$ , where  $i + j = i' + j'$ , bounds their count difference by  $V(i + j)$ . Let  $z_n = n - q_n$  count zeros. Then

$$p_n = kn - e z_n. \quad (7)$$

If  $\liminf z_n/n > 0$ , choose  $0 < \gamma \leq 1$  with  $z_n \geq \gamma n$  eventually; then  $p_n \leq (k - e\gamma)n$ , and the density shortcut applies. Assume  $\liminf z_n/n = 0$ . There are infinitely many  $\sigma$  with  $V(\sigma) \geq 2$ . Otherwise a sufficiently late tail would be balanced by the same endpoint comparison. Since zero recurs, it has a length- $L$  factor containing two zeros. Every length- $L$  factor would then contain a zero, giving  $z_n \geq n/L - O(1)$ , a contradiction.

If  $k = 2u + 1$ , fix  $\sigma_0$  with  $V(\sigma_0) \geq 2$ . For all large  $s$ , split the slots into  $u$  pairs of subtotal  $\sigma_0$  and a singleton carrying the rest. Independent variation gives width at least  $uV(\sigma_0) \geq k - 1$ , so the sweep applies.

For  $k = 2u$ ,  $u \geq 2$ , choose  $\sigma_0 < \sigma_1$  with  $V(\sigma_i) \geq 2$ , and put  $\Delta = \sigma_1 - \sigma_0$ . Suppose at a sufficiently large  $s$  the width is at most  $k - 2$ , and put  $\tau = s - (u - 1)\sigma_0$ . Using  $u - 1$  pairs of subtotal  $\sigma_0$  and one of subtotal  $\tau$  gives width at least  $k - 2 + V(\tau)$ , so  $V(\tau) = 0$ . Using  $u - 2$  pairs of subtotal  $\sigma_0$ , one of subtotal  $\sigma_1$ , and one of subtotal  $\tau - \Delta$  likewise gives  $V(\tau - \Delta) = 0$ . Now  $V(n) = 0$  implies  $q_i + q_{n-i}$  is constant. Successive differences give  $h_{i+1} = h_{n-i}$ , so the prefix of length  $n$  is a palindrome. The prefixes of lengths  $\tau$  and  $\tau - \Delta$  are therefore palindromes; the shorter is also a suffix of the longer, which consequently has period  $\Delta$ . Infinitely many width failures would give such periodic prefixes of unbounded length, forcing the whole word to have period  $\Delta$ . A period contains a zero, contradicting  $\liminf z_n/n = 0$ . Thus width is eventually at least  $k - 1$ , and the sweep finishes.  $\square$

## 4. THE UNIVERSAL LACUNARY CERTIFICATE

**Lemma 4.1.** *If every admissible walk is  $k$ -tail-covering, then any prescribed positive integer ratios  $R_n$  admit one increasing  $B$  with  $b_{n+1} \geq R_n b_n$  meeting  $kA$  for every admissible  $A$ .*

*Proof.* List pairs  $(q, r)$ ,  $q \geq 1$ ,  $0 \leq r < q$ , with each recurring infinitely often, by concatenating the finite lists with  $q \leq N$  for  $N = 1, 2, \dots$ . Write the list as  $(q_n, r_n)$ . Choose  $b_1 > 0$  in its prescribed class and let  $b_{n+1}$  be the least integer in the next class exceeding  $\max\{R_n b_n, b_n, n\}$ . The sequence grows as required. For any  $A$ , its class in (5) occurs infinitely often; the corresponding  $b_n$  are unbounded, so one exceeds  $X_0$  and belongs to  $kA$ .  $\square$

*Proof of Theorem 1.* Part (1) is Proposition 1.2. For  $d_2 \leq k$ , Proposition 3.1 and Lemma 4.1 give part (2).  $\square$

## APPENDIX A. PROOF OF THE FINITE INTERVAL LEMMA

Write  $u^m$  for the multiset of  $m$  copies of the positive integer  $u$ .

## A.1. Reduction to a dense triple.

**Lemma A.1** (Coprime rectangle). *If  $1 \leq p < q$  and  $\gcd(p, q) = 1$ , then*

$$\text{SS}(p^{q-1}, q^{p-1}) \supseteq [(p-1)(q-1), pq-1].$$

*The interval has  $p+q-1$  terms, at multiset cost  $p+q-2$ .*

*Proof.* For  $n$  in the interval choose  $0 \leq i < q$  with  $ip \equiv n \pmod{q}$ . Then  $j = (n - ip)/q$  satisfies  $-1 < j < p$ , so  $0 \leq j \leq p-1$ .  $\square$

**Lemma A.2** (Interlacing). *Let  $d, q \geq 1$ ,  $\gcd(d, q) = 1$ , and  $q \leq N$ . If  $[A, A+N-1] \subseteq \text{SS}(C)$ , then scaling  $C$  by  $d$  and adjoining  $d-1$  copies of  $q$  gives  $N$  consecutive subset sums.*

*Proof.* For  $0 \leq n < N$  choose  $0 \leq j < d$  with , and put  $t = ((d-1)q + n - jq)/d$ . Since  $0 \leq (d-1)q + n - jq < dN$ , we have  $0 \leq t < N$ . Thus

$$d(A+t) + jq = dA + (d-1)q + n$$

supplies the required interval.  $\square$

**Lemma A.3** (Two-generator targets). *For coprime  $1 \leq p < h$  and  $N \geq 2h$ , at most  $N - \lfloor N/h \rfloor$  terms from  $\{p, h\}$  have  $N$  consecutive subset sums.*

*Proof.* Start with the rectangle's interval of length  $p+h-1$ . Write  $N = sh + r$ ,  $s \geq 2$ ,  $0 \leq r < h$ , and add  $t = s-1 + \mathbf{1}_{\{r \geq p\}}$  copies of  $h$ . The translates overlap or abut and cover at least  $N$  integers, at cost  $h+p+s-3 + \mathbf{1}_{\{r \geq p\}}$ . At  $s=2$  its slack below  $N-s$  is  $h+r-p-1 - \mathbf{1}_{\{r \geq p\}} \geq 0$ ; increasing  $s$  increases slack by  $h-2$ .  $\square$

**Proposition A.4** (Inductive funnel). *Assuming SHARP below maximum  $M$ , it suffices at  $M$  to treat triples*

$$G = \{a < b < M\}, \quad \gcd(a, b) = \gcd(a, M) = \gcd(b, M) = 1, \quad a + b \geq M + 2. \quad (8)$$

*Proof.* Set  $H = G \setminus \{M\}$ ,  $d = \gcd(H)$ , and  $e = \max(H/d)$ . If  $|H| \geq 3$ , induction gives  $e$  consecutive subset sums at cost  $e-1$  from  $H/d$ . Adjoin  $\lceil M/e \rceil - 1$  copies of  $e$ , obtaining at least  $M$  consecutive sums. Scale by  $d$  and interlace with  $d-1$  copies of  $M$ ;  $\gcd(d, M) = 1$ . The cost is  $C = e + d + \lceil M/e \rceil - 3$ . Writing  $M = de + s$  with  $s \geq 1$  gives

$$M - 1 - C = (d-1)(e-2) + s - \lceil s/e \rceil \geq 0,$$

since  $e \geq 3$ . Otherwise  $H = \{a, b\}$ ,  $a < b$ . If  $d = \gcd(a, b) > 1$ , apply Lemma A.3 to  $\{a/d, b/d\}$  with target  $M$ , scale, and interlace with  $d-1$  copies of  $M$ . Here  $M > b$ ,  $d \geq 2$ , and  $\lfloor M/(b/d) \rfloor \geq d$ , so the cost is at most  $M-1$ . Now assume  $\gcd(a, b) = 1$ . If  $b \leq M/2$ , the same target lemma costs at most  $M-2$ . If  $b > M/2$  and  $a+b \leq M$ , one more copy of  $b$  extends the rectangle to length  $a+2b-1 \geq M$ , at cost at most  $M-1$ . If  $a+b = M+1$ , the rectangle itself suffices. Thus  $a+b \geq M+2$  remains. Finally, if  $d = \gcd(a, M) > 1$ , apply the target lemma to  $\{a/d, M/d\}$  with target  $M$ , at cost at most  $M-d$ , scale, and interlace with  $d-1$  copies of  $b$ ;  $\gcd(d, b) = 1$ . The case  $\gcd(b, M) > 1$  is symmetric.  $\square$

For a triple in (8), put

$$e = b - a, \quad h = M - b, \quad \mu = M - a = e + h. \quad (9)$$

Then

$$1 \leq h \leq a - 2, \quad (10)$$

so  $a \geq 3$  and  $\gcd(a, e) = \gcd(a, \mu) = 1$ .

### A.2. Paired movers.

**Lemma A.5** (Paired movers). *If  $3 \leq a < b < M$ ,  $\gcd(a, b) = 1$ , and  $b + M = ra$ , then for*

$$x = r - 1, \quad y = \lceil (M - r)/2 \rceil, \quad z = \lfloor (M - r)/2 \rfloor,$$

$\text{SS}(a^x, b^y, M^z)$  contains  $M$  consecutive integers and  $x + y + z = M - 1$ .

*Proof.* Here  $r \geq 3$ , and

$$M \geq a + r - 1, \quad M \leq a(r - 1) - 1 : \quad (11)$$

indeed  $2M \geq ra + 1 \geq 2a + 2r - 2$  and  $M = ra - b$ . Put  $p = \lfloor a/2 \rfloor$ ,  $q = \lfloor (a - 1)/2 \rfloor$ , and  $L = \max(pb, qM)$ . Then  $p + q = a - 1$ ,  $p \leq y$ , and  $q \leq z$ . For  $n \in [L, L + M - 1]$  choose  $w \in [0, a - 1]$  with  $wb \equiv n \pmod{a}$ . If  $w \leq p$ , use  $w$  unpaired copies of  $b$  and divide  $(n - wb)/a = i + r\kappa$ ,  $0 \leq i < r$ . Otherwise put  $w' = a - w \leq q$ , use  $w'$  copies of  $M$ , and divide  $(n - w'M)/a = i + r\kappa$ . The numerators are nonnegative, so  $\kappa \geq 0$ . Each of  $\kappa$  further pairs  $(b, M)$  contributes  $ra$ , and  $i \leq x$ . The copies are available provided

$$L + M - 1 + pM \leq a(ry + x), \quad (12)$$

$$L + M - 1 + qb \leq a(rz + x). \quad (13)$$

For example, in the first branch  $a(i + r\kappa) = n - wb$  and  $a(i + r(\kappa + w)) = n + wM$ ; these inequalities bound  $\kappa \leq z$  and  $\kappa + w \leq y$ . The second branch is symmetric. If  $L = qM$ , the inequalities reduce to  $aM - 1 \leq a(ry + x)$  and  $qra + M - 1 \leq a(rz + x)$ . The latter uses  $q \leq z$  and  $M - 1 \leq ax$ ; the former uses

$$ry + x \geq r(M - r)/2 + r - 1 \geq M,$$

since  $M \geq a + r - 1 \geq r + 2$ . If  $L = pb$ , the first inequality follows from  $p \leq y$  and  $M - 1 \leq ax$ . The second is  $(a - 1)b + M - 1 \leq a(rz + x)$ . Odd  $a$  gives  $p = q$  and  $qM > pb$ , so take  $a = 2v$ ,  $v \geq 2$ . Put  $\tau = M - vr = vr - b \geq 1$ . The slack is

$$D = 2vrz - 2(v - 1)b - 2v + 1 = 2vr(z - v + 1) + 2(v - 1)\tau - 2v + 1.$$

Here  $z = \lfloor (r(v - 1) + \tau)/2 \rfloor \geq v - 1$ . If  $z \geq v$ , then  $D \geq 2vr - 1 > 0$ . If  $z = v - 1$ , then  $\tau \neq 1$  (that would give  $z \geq v$ ), so  $\tau \geq 2$  and  $D \geq 2v - 3 > 0$ . Finally  $x + y + z = M - 1$ .  $\square$

### A.3. The symmetric residue path.

**Lemma A.6** (Centered frame). *If  $C$  has a subset sum in one common interval  $[L, U]$  for each residue modulo  $a$ , put  $x = \lceil (M - 1 + U - L)/a \rceil$ . Then  $\text{SS}(C \cup a^x)$  contains  $[U, L + ax]$ , with at least  $M$  integers.*

*Proof.* Subtract the chosen representative of  $n \pmod{a}$  from  $U \leq n \leq L + ax$ . The remainder is a nonnegative multiple of  $a$  at most  $ax$ .  $\square$

**Lemma A.7** (Symmetric  $\eta$  argument). *If  $a < b < M$  satisfy (10), are pairwise coprime, and  $a \nmid b + M$ , at most  $M - 1$  terms from  $\{a, b, M\}$  have  $M$  consecutive subset sums.*

*Proof.* Let  $p, q$  be units modulo  $a$ , let  $\eta \in [0, a - 1]$  represent  $qp^{-1}$ , and put

$$t = \min(\eta, a - \eta), \quad Z = \lfloor (a - 1)/t \rfloor, \quad r = a - 1 - tZ, \quad K = t - 1 + Z. \quad (14)$$

For  $(p, q) = (b, M)$  we have  $\eta \equiv \mu e^{-1} \pmod{a}$ . The values  $0, -1, 1$  would give respectively  $a \mid M$ ,  $a \mid b + M$ , and  $a \mid h$ , so  $2 \leq t \leq \lfloor a/2 \rfloor$ . If  $\eta = t$ , write  $0 \leq n < a$  as  $n = tk + j$ . The representatives  $jp + kq$  lie in  $[0, S]$ , where

$$S = \max\{(t - 1)p + (Z - 1)q, rp + Zq\}. \quad (15)$$

If  $\eta = a - t$ , write the consecutive residues  $t - a, \dots, t - 1$  as  $n = j - tk$ ; representatives lie in  $[0, S]$  with

$$S = (t - 1)p + Zq. \quad (16)$$

All are subset sums of  $p^{t-1}, q^Z$ . The centered frame with  $L = 0$  costs

$$\mathcal{B} = K + \lceil (M - 1 + S)/a \rceil. \quad (17)$$

We can also reverse  $(p, q)$ , since  $M$  is a unit modulo  $a$ . Always

$$K \leq \lceil a/2 \rceil. \quad (18)$$

Indeed, for  $c = \lceil a/2 \rceil$  the concave function  $t(c - t + 2)$  exceeds  $a - 1$  at both endpoints of  $[2, \lfloor a/2 \rfloor]$ , so  $\lfloor (a - 1)/t \rfloor \leq c - t + 1$ . Since  $S \leq KM$ , the budget holds whenever

$$(a - K - 1)M \geq a(K + 1) - 1. \quad (19)$$

For  $a = 3$  all unit residues are already excluded.

*Odd  $a = 2n + 1$ .* If  $K \leq n$ , the slack in (19) at  $K = n$  is  $n(\mu - 2) \geq 0$ . Furthermore  $t - 1 + \lfloor 2n/t \rfloor = n + 1$  forces  $t \in \{2, n\}$ , since  $(t - 2)(t - n) < 0$  for  $2 < t < n$ . Only  $\eta \in \{2, -2, n, n + 1\}$  remains. If  $\eta = 2$ , then  $e - h = \lambda a$  for  $\lambda \geq 0$  by (10). Here  $Z = n$ ,  $r = 0$ ,  $S = nM$ . With  $\epsilon = 0$  for  $h = 1$  and  $\epsilon = 1$  for  $h \geq 2$ , evaluation gives

$$(M - 1) - \mathcal{B} = n\lambda + h - 2 - \epsilon. \quad (20)$$

This is nonnegative if  $\lambda \geq 1$ , or if  $\lambda = 0$  and  $e = h = g \geq 3$ . For  $g = 1$ , take  $n + 1$  copies of  $a$ , one of  $a + 1$ , and  $n$  of  $a + 2$ . At selection level  $n + 1$ , offsets fill  $[0, a]$ ; at the next level offsets  $[1, a]$  remain. The intervals abut, of total length  $2a + 1 \geq a + 2$ . For  $g = 2$ , take  $n + 3$  copies of  $a$ , one of  $a + 2$ , and  $n$  of  $a + 4$ . With  $c = n + 1$ , each level  $\ell \in \{c, c + 1, c + 2\}$  contains  $\ell a + 2[0, a]$ . On  $[(c + 1)a, (c + 3)a]$ , the middle level supplies one parity and the other levels supply the other. The costs are respectively  $a + 1$  and  $a + 3$ , as required.

If  $\eta = -2$ , reverse  $b, M$ ; the reversed ratio is  $n$ . For  $n \geq 3$ , the normalized residues  $1, \dots, a$  have representatives

$$jM \ (1 \leq j \leq n - 1), \quad b + jM \ (0 \leq j \leq n - 1), \quad 2b, \quad M + 2b.$$

These use  $M^{n-1}, b^2$  and lie in  $[b, b + (n - 1)M]$ . The centered frame costs  $n + 1 + \lceil (nM - 1)/a \rceil \leq M - 1$ , since the slack is

$$a(M - n - 2) - (nM - 1) = (n + 1)\mu - 2n \geq 2.$$

For  $n = 2$ , representatives  $M, b, M + b, 2b, M + 2b$  lie in  $[b, M + 2b]$ . The cost condition is  $2e + 3h - 9 \geq 0$ . Since  $h \equiv 2e \pmod{5}$ , pairwise coprimality and  $1 \leq h \leq 3$  exclude  $e = 1, 2$ , so it holds.

For  $\eta = n$  or  $n + 1$  with  $n \geq 3$ , use the forward path in the first case and reverse in the second, making its ratio 2. In either case  $K = n + 1$  and  $S \leq (n - 1)b + M$ . The budget is at most  $a + 1 + \lceil ((n + 1)e + 2h - 1)/a \rceil$ , and its required slack is

$$ne + (a - 2)h - 2a + 1 \geq 0. \quad (21)$$

For  $h \geq 2$  this is at least  $n - 3$ . For  $h = 1$ , the first orientation gives  $a \mid 3e + 2$ , so  $e \geq 3$ ; the second gives  $a \mid e + 2$ , so  $e \geq a - 2$ . Both imply the inequality.

*Even  $a = 2n$ .* If  $K \leq n - 1$ , (19) holds, with slack  $n\mu + 1$  at  $K = n - 1$ . For  $a \geq 12$  and  $3 \leq t \leq n - 2$ ,  $t - 1 + \lfloor (2n - 1)/t \rfloor \leq n - 1$ . Thus maximal  $K = n$  requires  $t \in \{2, n - 1, n\}$ . As  $\eta$  is a unit, only even  $n$  and  $\eta = n \pm 1$  remain. For  $\eta = n - 1$ , (15) gives  $Z = 2$ ,  $r = 1$  and  $S = (n - 2)b + M$ . The slack is

$$ne + 2(n - 1)h - 2n + 1 \geq n - 1. \quad (22)$$

For  $\eta = n + 1$ ,  $h \equiv ne \pmod{2n}$ ;  $e$  is odd, so (10) forces  $h = n$ . Now  $S = (n - 2)b + 2M$  and the slack is

$$2n^2 - 7n + (n - 1)e + 1 \geq 2n(n - 3) > 0. \quad (23)$$

For  $a = 4, 6$  there are no unit residues beyond  $\pm 1$ . For  $a = 8$ ,  $\eta = 3$  gives  $S = b + 2M$  and slack  $8(M - 5) - (3M + b - 1) = 4e + 5h - 7 > 0$ . For  $a = 8$ ,  $\eta = 5$  forces  $h = 4$ ;  $S = 2b + 2M$  gives slack  $8(M - 5) - (3M + 2b - 1) = 3e + 5h - 15 > 0$ . For  $a = 10$ , reverse if necessary to make the ratio 3. The height is at most  $2b + 2M$ , and slack is  $10(M - 6) - (3M + 2b - 1) = 5e + 7h - 9 > 0$ . All cases succeed.  $\square$

*Proof of Theorem 2.2.* Use strong induction on  $M$ , with the statement vacuous for  $M < 3$ . Proposition A.4 leaves a triple satisfying (8). If  $a \mid b + M$ , use Lemma A.5; otherwise use Lemma A.7.  $\square$

## APPENDIX B. FORMAL VERIFICATION AND CORRESPONDENCE

The following parts of the new argument are checked in Lean. File paths are relative to `lean/Erdos1112Proof/`.

| Paper result                        | Lean declaration                        |
|-------------------------------------|-----------------------------------------|
| <b>Existence/Reciprocal.lean</b>    |                                         |
| Lacunary interpolation              | <code>reciprocal_interpolation</code>   |
| Existence with ratio $d_2 + 2$      | <code>existence_bound_reciprocal</code> |
| <b>Short/Intervals.lean</b>         |                                         |
| Coprime rectangle                   | <code>coprime_rectangle</code>          |
| Interlacing                         | <code>interlacing</code>                |
| Two-generator targets               | <code>two_generator_target</code>       |
| Centered frame                      | <code>centered_frame</code>             |
| <b>Short/Slots.lean</b>             |                                         |
| Slots and final-summand sweep       | <code>slots_from_run</code>             |
| <b>Short/Normalization.lean</b>     |                                         |
| Tail alphabet and gcd normalization | <code>normalized_walk</code>            |
| <b>Short/Binary.lean</b>            |                                         |
| Zero-liminf binary width argument   | <code>binary_eventual_width</code>      |
| Binary covering from zero liminf    | <code>binary_tail_covering</code>       |
| <b>Short/Funnel.lean</b>            |                                         |
| Inductive funnel                    | <code>sharpAt_of_funnel</code>          |
| <b>Short/Sharp.lean</b>             |                                         |
| Complete finite interval lemma      | <code>sharp_all</code>                  |

The reciprocal construction, binary alternative and complete finite interval lemma follow this paper in Lean. The old SHARP proof and certificate tables have been removed. The density dependency is still being formalized; the final non-existence theorem temporarily retains the earlier argument, whose verification does not certify the new Kneser shortcut. The definitions in `lean/Erdos1112.lean` allow repeated summands, phrase gap bounds additively, and include a proved equivalence between natural and integer ratio formulations. The repository records the current migration status and reproduction commands.

## REFERENCES

- [1] T. F. Bloom, *Erdős Problem #1112*, <https://www.erdosproblems.com/1112>.
- [2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique **28**, Université de Genève, Geneva, 1980.
- [3] P. Erdős, M. B. Nathanson, and A. Sárközy, *Sumsets containing infinite arithmetic progressions*, J. Number Theory **28** (1988), 159–166. [https://www.renyi.hu/~p\\_erdos/1988-33.pdf](https://www.renyi.hu/~p_erdos/1988-33.pdf)
- [4] M. Kneser, *Abschätzung der asymptotischen Dichte von Summenmengen*, Math. Z. **58** (1953), 459–484. <https://doi.org/10.1007/BF01174162>
- [5] J. Land, *A dichotomy for  $k$ -fold sumsets of bounded-gap sequences (Erdős Problem 1112)*, preprint, 2026. <https://doi.org/10.5281/zenodo.21568276>; source and Lean development: [https://github.com/beetree/math\\_erdos\\_1112](https://github.com/beetree/math_erdos_1112).